A pre-registered *in silico* falsification programme on a redox-switch model of parvalbumin-interneuron damage

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What was done. We built a mechanistic ODE model of oxidative damage to parvalbumin interneurons — a bistable peroxiredoxin/sulfiredoxin redox switch, a two-stage inducible defence ($\tau_m = 12\,\mathrm{h}$, $\tau_s = 63\,\mathrm{h}$), a Phillips–Robinson sleep–wake switch, and a Wilson–Cowan cortex closing the loop through disinhibition — and then spent most of the effort trying to break it. Every hypothesis was written as a numbered rule with its parts, and *every threshold was derived from expectations and variances before the computation ran.* Verdicts were recorded in an append-only ledger and were never revised afterwards; when a defect was found in a rule, the defect was recorded *beside* the verdict and the question was re-asked as a new rule on a fresh noise seed.

Of **127 pre-registered parts across 26 rules: 66 passed, 47 failed, 13 were unfalsifiable, and one fell outside its own analytic bracket.** More than a third of our own predictions were wrong. That ratio is the main result we wish to report, and the failures below were more informative than the successes.

What survived.

1. *Episode duration is set by defence memory, not by repair capacity.* Within a parameter set, 99.8 % of the variance of episode duration is explained by the history of oxidative load and 0.2 % by repair; the decomposition is exact by construction and the shares sum to unity to 10^{-16}. The dependence on defence memory is affine and does not saturate: a saturating form with a shared offset is rejected, although our quantitative out-of-fit prediction missed its own 2 SE band. Consequence: passively strengthening repair cannot reach the clinical duration target — it misses by a factor of 2 to 10.

2. *Slow defence memory generates a pre-episode precursor.* Removing it lowers the precursor in 107 of 174 sets, $p = 0.0015$. Slow load noise is not needed for this; its two independent justifications were withdrawn.

3. *The population lives in three phases, not two.* Between the healthy core and the collapsed core there is a band of cells that are *more* vulnerable than their collapsed neighbours and nevertheless intact — roughly 8.5 % of the population, seventeen standard errors from zero. It collapses to *exactly* zero when the slow defence is frozen, so it is a dynamical effect rather than a parametric one, and priming widens it without causing it. This is hormesis at the population level and it is a prediction of the model, not an artefact of counting.

4. *The rhythm breaks through a devil's staircase*, with rational rotation numbers, alternating cycle lengths on the fractional steps, and a cascade that adds one rather than doubling.

5. *Cell count enters only as effective spread of resilience*, not as numerical resolution; the $J \to \infty$ limit exists and was measured, with corrections converging as $1/J$.

6. *A capacity-dependent feedback loop converts frequent short episodes into rare long ones.* With per-cell capacity loss while damaged and slower recovery afterwards, mean episode duration rises from 13.1 to 25.6 days while the onset rate *falls* from 0.042 to 0.033 per at-risk set-day; episodicity is retained at 0.75 of the open model; the broken state is held after load is released, monotonically in release depth ($\rho = +1.000$, $p = 0.0083$, the attainable minimum); and progression exceeds a capacity-matched static control by +0.0231 against a minimum detectable effect of 0.0096 — replicated as +0.0223 and +0.0211 on two further independent noise seeds.

What failed.

- *A load-dependent feedback loop of the Grace type is refuted as a model of the disease.* Driving oxidative load with the damaged fraction does produce irreversibility, and it is not a mere gain knob: it latches states that no static load reaches at all. But irreversibility switches on at loop

strength 1.11 while episodes extinguish at 0.18 — a sixfold gap that is structural rather than marginal, and robust to a gentler breaking protocol. There is no window in which chronicity and episodicity coexist. The reason is now clear: load is a per-set quantity, so a loop acting through it moves the whole population together.

- *The capacity loop changes the level and the duration of damage but not the shape of risk in time.* Its hazard trend exceeds a capacity-matched static arm by only 1.94 against a threshold of 2.23 computed exactly for that comparison.
- *Bursty load does not beat smooth load*, contrary to the hypothesis that rare strong excursions dominate; and the measured quantity that first appeared to support it turned out to saturate.
- Several precursor-slope predictions were *unfalsifiable* at the grid resolution we had chosen: the minimum detectable effect was larger than the threshold we had written down.

What the failures taught (the transferable part). The single most useful finding is negative and methodological:

In a model with absorbing states and a heterogeneous population, no absolute null exists for any accumulation measure. The damaged fraction rises on its own, because a bistable switch does not release a collapsed set until load is lowered; and the hazard of new episodes declines on its own, because the most vulnerable cells are absorbed first and the survivors are less vulnerable. Two of our own controls asserted that these baselines were zero. Both were false, and — this is the instructive part — both *passed at small sample size and failed as data accumulated*, because the baseline is fixed while the detection threshold shrinks as $1/\sqrt{n}$. A control whose null is false is passed by lack of power. Every claim must therefore be a matched difference between arms brought to the same damage level, never a test against zero.

Nine further statistical traps were catalogued, each of which had once produced a false conclusion in our own hands and each of which is now closed by a check in the code. Four are worth naming: a variance-ratio statistic used in four rules was shown analytically to equal $1 + \mathrm{CV}^2$ of the underlying rates and was declared unfit; an equivalence claim whose tolerance exceeds half the range of its own curve cannot be falsified; a threshold must be *attainable*, since $p \leq 0.05$ on an exact permutation test over three points has a floor of $1/3! = 0.167$, and we once recorded "failed" for a perfectly monotone result; and a comparison point chosen as the nearest element of a grid that does not bracket the target answers a question about the grid's edge.

Scope, stated as a result rather than a caveat. The model has no absorption-free stationary state, so its domain of validity is windows short compared with the accumulation time. It is therefore a model of the *generation and termination of an episode* under oxidative load, not of the course of the illness. We did not attempt the global picture and do not claim it. The capacity loop is the only mechanism we tested that makes progression appear at all, and it does so by lengthening episodes rather than by making them more frequent.

Reproducibility. A fused compiled kernel giving a 3.3-fold wall-clock speed-up was proved identical to the reference integrator at two levels: bitwise over 67 output fields on nine parameter pins (worst deviation 9.5×10^{-15} against a tolerance of 10^{-12}), and over the entire set of verdicts. Block results are cached under a key that includes the hash of the model source, the kernel included — a discipline added after a kernel defect was found that no pin touched. Step-size convergence separates two classes of quantity: continuous means converge to within 0.5 % by $M = 800$, whereas counting quantities do not, with a Richardson limit 2.0 % above the value at $M = 1600$; counting quantities are therefore not tested against thresholds finer than that. Whole project could be found in the github.

What we would ask next. The load loop failed because load is a per-set quantity, so a loop acting through it moves the whole population together; a structure in which regions couple through narrow bottlenecks would break that homogeneity, and is the natural next test. The capacity loop lengthens episodes but leaves the shape of risk unchanged, which suggests that progression on the scale of the illness needs a component that does *not* fully return. We have deliberately not added it: a mechanism introduced to produce the result it was designed to produce is not a test, and the rule that would falsify it has to be written first.